# Resources
## REAL VARIABLES I
## Fall 2026

For every topic we cover, I will provide either lecture notes or a precise reading reference. The books below offer complementary treatments; you do not need to purchase them.

## 0.1 Main textbook

- Luigi Ambrosio, Giuseppe Da Prato, and Andrea Mennucci, *Introduction to Measure Theory and Integration*, Publications of the Scuola Normale Superiore 10, Edizioni della Normale, 2011. This is our principal reference. Publisher page and electronic access.

## 0.2 Additional references

- Elias M. Stein and Rami Shakarchi, *Real Analysis: Measure Theory, Integration, and Hilbert Spaces*, Princeton Lectures in Analysis III, Princeton University Press, 2005. Book record.
- Sheldon Axler, *Measure, Integration & Real Analysis*, Graduate Texts in Mathematics 282, Springer, 2020. The author provides a legal, free electronic edition. Open-access book.
- Gerald B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999. This is a more advanced reference with extensive coverage beyond the course. Publisher page.

## 0.3 Freely available notes and texts

The following materials are available online and are also represented in the course `books/` archive. They are optional supplementary reading, not required purchases.

- Jeff Viaclovsky, *18.125 Measure and Integration* (MIT OpenCourseWare, Fall 2003). A 24-part set of lecture notes. Online lecture notes.
- Brian S. Thomson, Judith B. Bruckner, and Andrew M. Bruckner, *Real Analysis*, 2nd ed., 2008. Free PDF.
- Steve Cheng, *A Crash Course on the Lebesgue Integral and Measure Theory.* Online PDF.
- D. H. Fremlin, *Measure Theory*, Volumes 1–5. The first two volumes are especially suitable as introductory references. Online contents and PDFs.
- Terence Tao, *An Introduction to Measure Theory.* Author's preliminary PDF.
- John K. Hunter, *Measure Theory.* Online PDF.
- John E. Hutchinson, *Analysis 2, 2012: Lecture Notes.* Online PDF.
- Jeffrey Steif, *Lecture Notes on Measure Theory, Integration Theory and Differentiation Theory.* Online PDF.

## 0.4 Office hours

Office hours are part of the course. Bring questions about definitions, proofs, examples, reading, or homework - even when your question is only ”where should I start?”
Discussing homework in office hours is encouraged. If you give a clear, complete solution of a designated homework exercise, I may propose to you a grade for that exercise during the meeting. If you are satisfied with the grade, you will not need to submit a written solution for that exercise.

## 0.5 Access and well-being

- Center for Educational Access
- Counseling and Psychological Services
- U of A Cares
- University emergency information
- RazALERT